Remote Sensing of Environment 238 (2020) 111017

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Characterizing land cover/land use from multiple years of Landsat and MODIS time series: A novel approach using land surface phenology modeling and random forest classifier

Lan H. Nguyen[a], Deepak R. Joshi[b], David E. Clay[b], Geoffrey M. Henebry[a,c,*]

[a] Geospatial Sciences Center of Excellence, South Dakota State University, Brookings, SD 57007, USA
[b] Department of Agronomy, Horticulture & Plant Science, South Dakota State University, Brookings, SD 57007, USA
[c] Department of Natural Resource Management, South Dakota State University, Brookings, SD 57007, USA

ARTICLE INFO

Keywords:
Land cover/land use
Land surface phenology
Random forest classifier
Croplands
Grasslands
South Dakota

ABSTRACT

Over the last 20 years, substantial amounts of grassland have been converted to other land uses in the Northern Great Plains. Most of land cover/land use (LCLU) assessments in this region have been based on the U.S. Department of Agriculture - Cropland Data Layer (USDA - CDL), which may be inconsistent. Here, we demonstrate an approach to map land cover utilizing multi-temporal Earth Observation data from Landsat and MODIS. We first built an annual time series of accumulated growing degree-days (AGDD) from MODIS 8-day composites of land surface temperatures. Using the Enhanced Vegetation Index (EVI) derived from Landsat Collection 1's surface reflectance, we then fit at each pixel a downward convex quadratic model to each year's progression of AGDD (*i.e.*, $EVI = \alpha + \beta \times AGDD - \gamma \times AGDD^2$). Phenological metrics derived from fitted model and the goodness of fit then are submitted to a random forest classifier (RFC) to characterize LCLU for four sample counties in South Dakota in three years (2006, 2012, 2014) when reference point datasets are available for training and validation. To examine the sensitivity of the RFC to sample size and design, we performed classifications under different sample selection scenarios. The results indicate that our proposed method accurately mapped major crops in the study area but showed limited accuracy for non-vegetated land covers. Although all RFC models exhibit high accuracy, estimated land cover areas from alternative models could vary widely, suggesting the need for a careful examination of model stability in any future land cover supervised classification study. Among all sampling designs, the "same distribution" models (proportional distribution of the sample is like proportional distribution of the population) tend to yield best land cover prediction. RFC used only the most eight important variables (*e.g.*, three fitted parameter coefficients [α, β, and γ]; maximum modeled EVI; AGDD at maximum modeled EVI; the number of observations used to fit CxQ model; and the number of valid observations) have slightly higher accuracy compared to those using all variables. By summarizing annual image time series through land surface phenology modeling, LCLU classification can embrace both seasonality and inter-annual variability, thereby increasing the accuracy of LCLU change detection.

1. Introduction

Since 2000, the US Northern Great Plains region (NGP, embraces five U.S. States: North Dakota, South Dakota, Nebraska, Minnesota, and Iowa) has experienced substantial land cover/land use (LCLU) change, especially the grassland conversion to cropland (Faber et al., 2012; Johnston, 2014; Wright and Wimberly, 2013a), due to contribution of many factors including government policy (Lark et al., 2015), an aging workforce of farmers, commodity prices (Claassen et al., 2011), energy development (Preston and Kim, 2016; Singh et al., 2017), and rental agreements. Johnston (2013) estimated wetlands losses of 5000–6000 ha per year over the past decade due to cropland expansion. Wright and Wimberly (2013b) found a net loss of 530,000 ha in grass-dominated lands from 2006 to 2011 in the Western Corn Belt (area of the Midwest U.S. dominated by agriculture, mostly corn and soybean cultivation) at the rate of 1.0% to 5.4% annually. Those changes have caused habitat fragmentation and threaten the loss of regional biodiversity (Stephens et al., 2008; Meehan et al., 2010; Mutter et al., 2015;

* Corresponding author at: Department of Geography, Environment, and Spatial Sciences, Center for Global Change and Earth Observations, Michigan State University, East Lansing, MI 48824, USA.
E-mail address: henebryg@msu.edu (G.M. Henebry).

https://doi.org/10.1016/j.rse.2018.12.016
Received 4 July 2018; Received in revised form 23 November 2018; Accepted 10 December 2018
Available online 17 December 2018
0034-4257/ © 2018 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/BY-NC-ND/4.0/).

Otto et al., 2016; Wimberly et al., 2018), increased soil erosion and water pollution (Vaché et al., 2002; Montgomery, 2007), and increased net carbon debt (Fargione et al., 2009; Searchinger et al., 2008).

Reliable data sources to quantify land cover changes come from the periodic statistical surveys by the USDA, such as the National Resources Inventory (NRI) by the Natural Resources Conservation Service (NRCS), and Quick Stats Database by the National Agricultural Statistics Service (NASS). However, changes detected from those datasets lack sufficient spatial and temporal details preventing identification of where and when transitions occur (Claassen et al., 2011; Rashford et al., 2011). These limitations prohibit the use of NRI and NASS data for comprehensive analysis of LCLU change. Thus, to better understand land cover changes, researchers have turned to geospatial datasets such as the National Wetland Inventory, the National Land Cover Datasets (NLCD), and, especially, the USDA Cropland Data Layer (CDL) due to finer spatiotemporal resolution and high number of classes (Faber et al., 2012; Wright and Wimberly, 2013a, 2013b; Johnston, 2013, 2014; Lark et al., 2015; WWF, 2016). The relatively high spatiotemporal resolution of the CDL enables tracking of not only land cover transitions but also crop rotations on a field basis. Use of the CDL can be justified by overall accuracy of 85% to 95% for major crops, and often 97% producer and user accuracies for corn and soybean (Boryan et al., 2011). However, recent studies have raised concerns about CDL-based analysis, especially when using a simple bi-temporal comparison approach with at least one study period before 2010 (Reitsma et al., 2016; Lark et al., 2017). It is important to aware that the CDL is meant to monitor agricultural land cover annually rather than tracking changes over time; and it has undergone substantial methodological changes over time (Mueller and Seffrin, 2006; Boryan et al., 2011). Independent validation of the 2006 and 2012 South Dakota CDL (Reitsma et al., 2016) revealed inconsistencies in the accuracy of generalized cropland and grassland classes. Reitsma et al. (2016) reported that the producer and user accuracies were dependent on location and land use. For example, in southeastern South Dakota, the producer accuracies for cropland and grassland were 88.4% and 38.9%, respectively. The importance of land cover knowledge raises the need for a more consistent land cover/land use product.

It is worth to note that the conventional classification methods, such as ones first applied to the CDL and the more general NLCD, were developed in an era of data scarcity and limited computational power. These methods focused on comparing just a few scenes and were challenged by classification error between spectrally-similar covers. Additionally, in areas with frequent morning cloud cover, collecting even a few cloud-free scenes over a year could be difficult, if not impossible. In the current era, a rapid increase of accessible Earth Observation data (particularly from the Landsat archive and its augmentation by newer sensors, such as Sentinel 2A and 2B) coupled with improved computing and storage capability is leading to the emergence of methods for generation annual land cover products from time series data as multi-temporal classification has been proved as superior to classification relying on just a few scenes (Franklin et al., 2015).

Shifts in land cover or environmental conditions, management practice, disturbance may lead to a spatiotemporal variation of land surface phenology (LSP), *i.e.*, seasonal patterns of reflectance from the vegetated land surface as observed using remote sensing (Henebry and de Beurs, 2013). Several methods have been developed to detect LSP that estimate timings of phenophase transitions during the growing season based either on pre-defined thresholds of vegetation indices (Lloyd, 1990; Reed et al., 1994; White et al., 1997) or on features of fitted curves (Zhang et al., 2003; de Beurs and Henebry, 2004). Among many proposed LSP models, the logistic curve (and its modifications) is one of the more commonly used fitting methods. Application of the logistic curve seems more suitable to landscapes dominated by woody vegetation (Zhang et al., 2003; Beck et al., 2006; Ahl et al., 2006; Baumann et al., 2017). Characterizing phenology can be more difficult for herbaceous than for woody vegetation due to strong interannual variation in grasslands and croplands (Schwartz and Reed, 1999). Another LSP fitting approach originates from traditional phenological models that relate to the progression of thermal time during the growing season to events in plant development. Studies have shown that the temporal development of remotely sensed vegetation indices in temperate ecosystems can be well approximated as a quadratic function of accumulated growing degree-days (AGDD) because AGDD captures well the seasonal course of insolation at mid-latitudes (de Beurs and Henebry, 2004; Henebry and de Beurs, 2013; Krehbiel and Henebry, 2016; Krehbiel et al., 2016, 2017).

LSP-based classifications that use time series data may be more accurate than traditional classification methods. Phenological time series can provide a larger number of sequentially related predictor variables to be exploited by machine learning classifiers. Over the past decade, LCLU change studies have included LSP information to map vegetated land cover and changes at regional to global scales (de Beurs and Henebry, 2004; Friedl et al., 2010; Gu et al., 2010; Zhong et al., 2011; Clerici et al., 2012; Zhu and Woodcock, 2014; Krehbiel et al., 2016; Qader et al., 2016).

A downward convex quadratic (CxQ) LSP model linking a vegetation index (such as the NDVI) to thermal time as measured by AGDD was first applied to compare spring green-up dynamics before and after the collapse of the Soviet Union (de Beurs and Henebry, 2004) to detect land cover change in a noisy AVHRR time series. Using a similar approach with Landsat time series for NDVI and MODIS for AGDD, Krehbiel et al. (2016, 2017) characterized the conversion of croplands to residential areas near Omaha, NE and Minneapolis-St. Paul, MN. Although de Beurs and Henebry (2004) and Krehbiel et al. (2016, 2017) successfully detected land cover changes in their study area through significant shifts in the value of phenometrics, neither studies quantified areal changes and generate spatially explicit LCLU change maps. On the other hand, Zhong et al. (2011) differentiated crop types in San Joaquin Valley, CA with an overall accuracy of above 70% for all three study years using only phenological metrics derived from MODIS NDVI time series. Using a very similar approach, Qader et al. (2016) accurately (generally > 90%) mapped broad dominant vegetation cover classes (cropland, grassland, and shrubland) in Iraq and characterized changes from 2002 to 2012. Compared to Zhong et al. (2011), Qader et al. (2016) sacrificed the classification detail by fewer classes for higher accuracy. However, both land cover change maps were produced at 250 m using MODIS NDVI data as input and neither used thermal data.

Understanding rapid land change is not only critical for management and conservation of prairie habitats and ecosystem services, but also for projecting production of crops and biofuels and the impacts of land conversion on rural infrastructure, such as roads and water quality. Hence, there is a need for a LCLU dataset with good spatiotemporal coverage as well as consistent accuracy through time. Taking advantage of abundant satellite observations, we developed a novel approach to mapping LCLU using as input only information derived from LSP modeling. First, we fitted a downward CxQ LSP model at each pixel using Landsat EVI and MODIS AGDD co-registered time series. Through the model fitting, we described phenological characteristics of each land cover type (timing of the growing season, peak EVI, timing of the peak EVI, *etc.*) as a simple functional form to allow cross-comparison. Fitted parameter coefficients and phenometrics at each pixel are then submitted to a random forest classifier (RFC) to generate a map depicting LCLU. We applied this approach to representative cropland areas in South Dakota for three years (2006, 2012, 2014) that reference point datasets are available for training and validation. Our objective is to examine how well we can classify land cover using only information from the land surface phenology modeling process. By utilizing the entire annual time series, we have multiple views of the surface that should help to better characterize LSP and thereby improve the classification.

We started the classification with three broad categories:

"cropland", "grassland", and "others" as used by the reference point datasets. Previous studies have suggested that classification by machine learning algorithms are sensitive to a sample size and design (*e.g.*, amount of each land cover type in the sample dataset) (Jin et al., 2014; Colditz, 2015; Millard and Richardson, 2015; Lyons et al., 2018). Accordingly, we generated multiple RFC models for each county-year using different sample datasets constructed from the CDL and the reference points to understand in which scenarios do RFC models perform best. We first examined how RFC models respond to increasing sample size with no control in sampling design. Next, RFC performances were evaluated using one of the following controls: (1) the proportional distribution among classes in the sample dataset was equivalent to the proportional distribution found in the re-classified CDL ("same distribution"), or (2) each class in the sample dataset contained the same number of pixels ("same size"). We also examined the effect of variable importance on RFC performance. Lastly, we demonstrate the capability of LSP-based classification to map major crops in two representative counties in South Dakota: Roberts and Codington.

2. Study area and data

2.1. Study area

Among the States composing the NGP region, we selected South Dakota as our study area for three reasons: (1) it has continental climate with high seasonal and interannual variation, pronounced gradients in precipitation (east-west) and temperature (north-south); (2) the State has strong livestock, dairy, and row crop production industries that make it economically viable to convert grasslands to croplands (Reitsma et al., 2015); and (3) unique high resolution reference datasets were available for training and validation.

Initially, we intended to focus on all 13 counties in South Dakota that have at least 90% of county area within overlap zones of Landsat paths (grey counties in Fig. 1) to maximize the potential number of cloud-free Landsat observations. However, due to limitations in the training dataset, we restricted our focus to four counties: Bon Homme, Codington, Roberts, and Walworth. In each of these four focal counties, we characterized LCLU for three years (2006, 2012, 2014) separately. According to 2014 CDL data, cropland is a dominant land cover in Roberts, Codington and Bon Homme, accounting for approximately 56%, 56% and 67% of county area, respectively (Figs. S1–S4). Major crops in these counties include corn, soybean, spring wheat (in Roberts and Codington), winter wheat (in Bon Homme), and alfalfa. > 80% of croplands were used for corn and soybean production. On the other hand, grassland is the predominant land cover in Walworth, covering 49% of the county area. Croplands, which consist mostly of corn/soybean, sunflower and winter wheat, cover 41% of the county.

2.2. Data

2.2.1. Reference point dataset for training

In each NASS region of South Dakota, 1600 random points were generated (Reitsma et al., 2015 and unpublished data) and laid over high-resolution imagery obtained from the National Agricultural Imagery Program (NAIP; USDA-FSA, 2013). The NAIP data were collected during the growing season at 2-meter resolution for 2006 and 1-meter resolution for 2012 and 2014. Information from three visible bands was used to construct a natural color image. Based on various features presented in these high-resolution images, an analyst classified the dominant land use in 2006, 2012, and 2014 at each location into one of five broad categories—cropland, grassland, habitat, not-agriculture, or water. A total of 43,200 points (1600 points in each NASS region × 9 regions × 3 years) were manually classified in South Dakota across the three years. For field validation, we randomly selected 100 points across the entire state for ground-level observations. The image-based classification and field observations matched in every case, giving us high confidence in the accuracy of the image-based classification.

There are 270 points in Roberts, 172 points in Codington, 122 points in Walworth, and 157 points in Bon Homme (see Figs. S1–S4 for maps of point locations). Although the original reference dataset consisted of five broad types, we are only interested in the croplands and grasslands; thus, we regrouped the reference dataset into just three categories: "cropland", "grassland", and "others".

2.2.2. Cropland Data Layer

The USDA Cropland Data Layer is a crop-specific land cover raster created annually for the continental United States by the NASS using moderate resolution satellite imagery and extensive agricultural ground observations (https://nassgeodata.gmu.edu/CropScape/). The CDL, which was first produced in 1997 for North Dakota, has covered the entire Great Plains yearly since 2006, with approximately 130 classes and a spatial resolution of 30 m at best (Boryan et al., 2011). For this study, we regrouped the CDL land covers into three classes to match the reference dataset. "Cropland" category contains the CDL's field crops

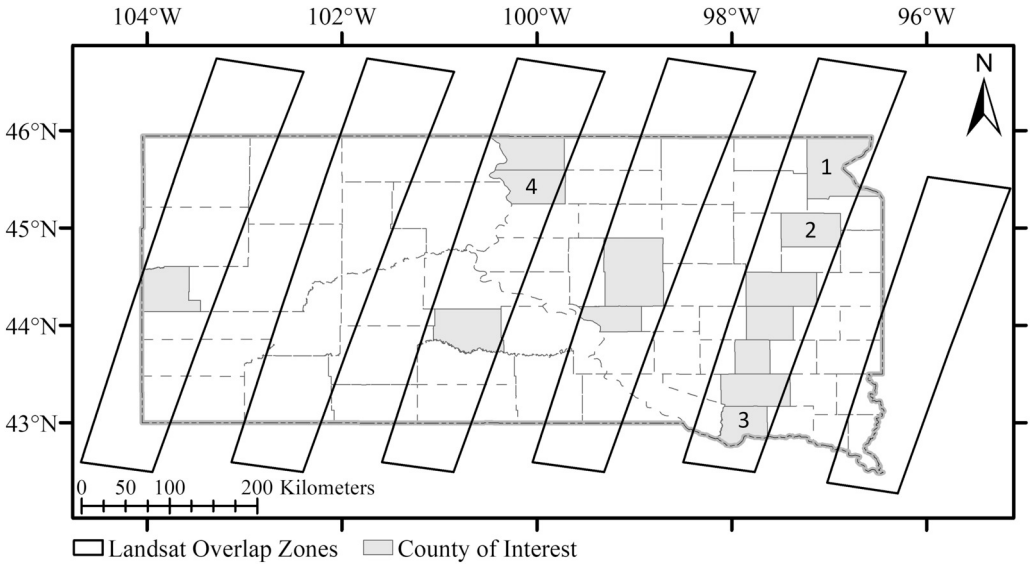


Fig. 1. South Dakota counties that fall mostly (> 90%) within Landsat overlap zones are indicated in grey. Four counties (1-Roberts, 2-Codington, 3-Bon Homme, and 4-Walworth) are the focus of this investigation.

(including alfalfa), vegetables, and tree fruits. All grass-like classes, including native grassland, pasture/hay, other hay/non-alfalfa, were combined to create a broad "grassland" category. The remaining CDL classes were put into the "others" category. Crosswalk details appear in Table S1.

2.2.3. Landsat Collection-1 surface reflectance product

In 2016, the U.S. Geological Survey (USGS) implemented the new organizational method for the Landsat archive called Collections to ensure consistent quality through time and across instruments for Landsat Level-1 products (USGS, 2016). This data structure provides a consistent archive of known data quality to support time-series analyses and data "stacking". Collection 1 initially consists of Level-1 products generated from Landsat 4–5 TM, Landsat 7 ETM+, and Landsat 8 OLI/TIRS instruments that are assigned to one of three categories: Tier-1, Tier-2, or Real-Time. Here, we used only the surface reflectance product from Landsat scenes in Tier-1 (WRS-2 Paths: 28–33 & Rows: 29–34), because those images are produced with the highest quality data available and, thus, are better suited for time series analysis and modeling.

2.2.4. MODIS Land Surface Temperature (LST) product

We used the Collection 5 MODIS level-3 global Land Surface Temperature and Emissivity 8-day composite products at 1000 m resolution from both Aqua (MYD11A2) and Terra (MOD11A2) satellites (NASA LP-DAAC, 2013). Both the "LST_Day_1km" and "LST_Night_1km" scientific datasets from the M{O|Y}D11A2 products were used. Overpass time for daytime (nighttime) observations are about 1030 (2230) local solar time for Terra and about 1330 (0130) local solar time for Aqua. The MODIS LST are provided in a sinusoidal grid format and display the mean clear-sky LST in Kelvin observed during an 8-day time frame. To use these LST products along with Landsat data, we re-projected the MODIS products to Albers Equal Area Conic projection, and then resampled the layer to 30 m using bilinear interpolation. The LST time series was converted from Kelvin to degrees Celsius for calculation of the thermal time used in the LSP modeling.

3. Methods

3.1. Construction of Landsat EVI and MODIS AGDD time series

The EVI was calculated from Landsat Collection-1 surface reflectance product (courtesy of the U.S. Geological Survey) as described in the Landsat Spectral Indices Product Guide (USGS, 2017). The outputs were re-projected into Albers Equal Area Conic projection to match with other datasets (MODIS LST, CDL). We then applied cloud/snow masks delivered with the product to remove "bad" observations (snow, high confidence cloud, or cloud shadow pixels). EVI values outside the valid range (from 0 to 1) were also excluded. The remaining "good" EVI values at each pixel were then stacked in chronological order from the first day of the year (DOY = 1) to the final day of the year (DOY = 365, or 366 in leap years). A number of valid observations was significantly lower in 2012 compared to 2006 and 2014 due to the end of Landsat 5 TM sensor operations.

The AGDD were calculated from MODIS LST as follow:

$$GDD_t = \max\left\{\frac{T_{max,t} + T_{min,t}}{2}, 0\right\} \quad (1)$$

$$AGDD_t = AGDD_{t-1} + 8 \times GDD_t \quad (2)$$

where GDD_t are the growing degree-days for compositing period (t), $T_{max,t}$ and $T_{min,t}$ are the highest and lowest LST values from available MODIS observations during the compositing period. Accumulation of GDD starts at the beginning of the calendar year and continues to the end of the year, but the temporal resolution of the LST composites is 8 d. Thus, we multiply the composite-calculated GDD by 8 to rescale it.

While good EVI observations from Landsat may occur at any DOY, there are only 46 MODIS composites each year. Although each pixel of the MODIS LST composite contains the best possible estimate during an 8 d period, each composite is linked to a single DOY, starting from DOY 1 to DOY 361 at 8 d intervals. Here we linked each EVI value from a specific DOY to the AGDD value associated with the "nearest" compositing period. For example, EVI of DOYs from 11 to 14 were coupled with AGDD composite 3 which is tied to DOY 17. If EVI was obtained right in the middle of the two MODIS compositing periods, paired AGDD is calculated as the average of the two AGDD compositing periods. For example, EVI of DOY 5 is paired with mean AGDD of composite 1 and 2 (DOY 1 and DOY 9). A complete list of possible EVI × AGDD pairs in a year appears in Table S2. As multiple EVI values may be linked to a specific AGDD compositing period, only the highest EVI of each available DOY will be used for model fitting.

3.2. Convex quadratic (CxQ) model for land surface phenology

The CxQ model for land surface phenology first linked the NDVI to AGDD (de Beurs and Henebry, 2004; Henebry and de Beurs, 2013; Krehbiel et al., 2016, 2017). Here, we used the EVI—Enhanced Vegetation Index (Huete et al., 2002)—for LSP modeling:

$$EVI = \alpha + \beta \times AGDD - \gamma \times AGDD^2 \quad (3)$$

where α, β, γ (alpha, beta, and gamma, respectively) are the parameter coefficients to be fitted. The negative sign on gamma in Eq. (3) is to show that we seek a fitted curve that is downward arching, since the EVI values will go up and then down over the growing season. Only a negative gamma coefficient will produce this shape and, thus, we retained only fits with a negative gamma coefficient. Occasionally, we may retrieve model fits with positive gamma coefficient for non-vegetated pixels (developed or barren areas). However, those fits were both very few and were deemed failed model fits.

Using the EVI and AGDD time series, we fitted the CxQ model at each pixel over the growing season in each year as described in Eq. (3) (Fig. 2). To identify the breadth of the fitting window of the growing season within year, we first calculated the change in EVI (ΔEVI) and AGDD (ΔAGDD) at each step and the rate-of-change from these two variables (ΔEVI/ΔAGDD, which measures how quickly EVI changes as a function of thermal time). We then looked for two transition points (Fig. 2a). Our approach to detecting the two transition points originates from Zhang et al. (2003), where MODIS EVI time series were fitted as a double logistic function of time. On the DOY axis, the two transition points are those with the highest rate-of-change in EVI (ΔEVI/Δt, which measures how quickly EVI changes as a function of calendar time). If data points are distributed equally on the time (t) axis (Δt remains unchanged), we expect to have extreme rates-of-change (the two transition points) around the maturity phase (highest EVI increasing rate) and the senescence phase (highest EVI decreasing rate) of vegetation as absolute changes in EVI (ΔEVI) are highest around these times. In our case, Landsat observations were not equally distributed along the thermal time axis and vary each year. In addition, we calculated the rate-of-change as ΔEVI/ΔAGDD. As the thermal time axis (AGDD) was used instead of the DOY axis, our calculated rates-of-change can be very high at the beginning or the end of the year due to small changes in temperature. Therefore, we cannot use only the ΔEVI or the ΔEVI/ΔAGDD values to detect transition points. Here, we coupled the rate-of-change with EVI change to detect the transition points. We first identified two highest and two lowest values of ΔEVI/ΔAGDD (Fig. 2b). Then, among the two highest rates-of-change, the one with maximum ΔEVI (largest increase in EVI) was labeled as the first transition point. Similarly, between the two lowest rates-of-change, the one with minimum ΔEVI (largest decrease in EVI) was labeled as the second transition point (Fig. 2c). As shown in Fig. 2b, the lowest point is not the one selected as that point does not have high ΔEVI (high ΔEVI/ΔAGDD values are due to small ΔAGDD). The two transition points

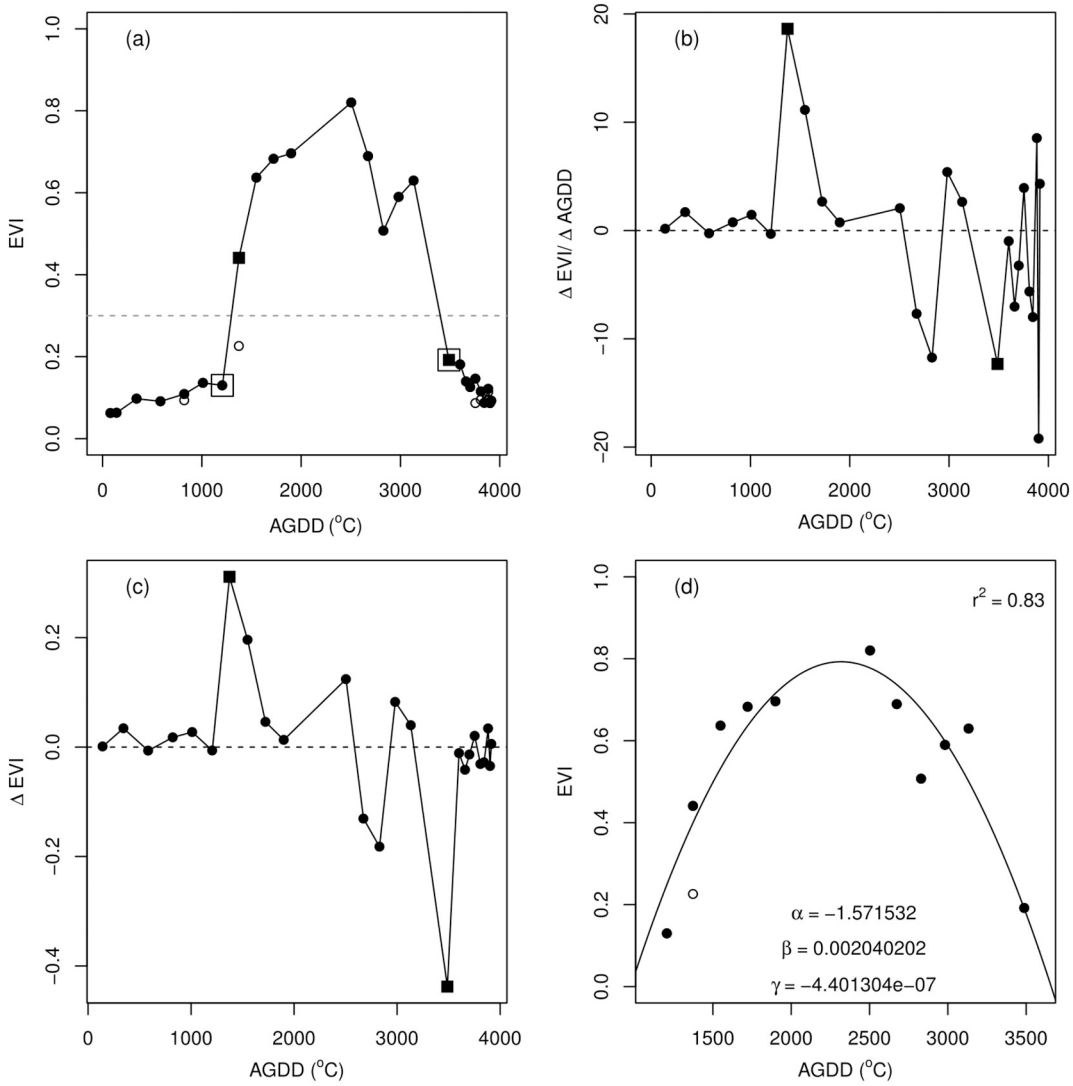


Fig. 2. Fitting the CxQ model to EVI and AGDD time series for a 2006 sample corn pixel in Roberts County (N45.8090, W96.9028): (a) EVI and AGDD time series, (b) rate-of-change (ΔEVI/ΔAGDD) and (c) EVI changes (ΔEVI) along the thermal time axis, and (d) the fitted CxQ curve. In (b) and (c), ΔEVI and ΔEVI/ΔAGDD between two days were paired with AGDD value of the later day. The two transitions are displayed as filled squares (a, b, c) and the fitting window is in between the two unfilled square (a). The model was fitted only for valid observations (filled circles).

consist of one point with highest ΔEVI/ΔAGDD value (Fig. 2b) and one point with second lowest ΔEVI/ΔAGDD but lowest in ΔEVI (Fig. 2b and c). From the two transition points, we searched outward to find two Landsat observations nearest in thermal time that fall below a certain EVI threshold (Fig. 2a). If the highest observed EVI (y_{max}) is above 0.65, the EVI threshold is 0.3. The EVI threshold is 0.2 for y_{max} ranging from 0.4 to 0.65; and there is no cut-off threshold if y_{max} is smaller than 0.40. In theory, only three points are required to retrieve the unique solution for a quadratic fit. However, to achieve a better fitted model, we restricted the fitting to pixels with at least five valid data points per year. Hence, we failed to produce proper quadratic fits for many pixels, mostly non-vegetated surfaces, and in the Scan Line Corrector (SLC) data gaps of Landsat 7 (*cf.* maps for 2012 in Figs. 3 and S12–S14). Those pixels were then flagged as no-data.

From each fitted model, we derived a suite of 16 metrics including fitted parameter coefficients, derived phenometrics, and model fit statistics (Table 1) that describe the shape and location of the fitted curve in the EVI × AGDD space, the goodness of model fit, and other relevant characteristics. These metrics, which were distilled from the pixel time series for each year, were then used as input for the LCLU classification task.

3.3. Land cover/land use classification using a random forest classifier (RFC)

3.3.1. Blended sample dataset

To increase the size of sample data, we blended reference points with the CDL. Note that overall accuracy of the CDL can be low at some times and places. For example, overall accuracy of crop classes (the most accurate classes) in South Dakota are only 61.2% in 2006, 74.7% in 2012, and 86.8% in 2014. To increase accuracy of sample data selected from the CDL, we first extracted core patches from the CDL, *i.e.*, center pixels surrounded by eight pixels (Queen's neighborhood) of the same type, to reduce misclassification that happens more frequently at the edge. We then compared LCLU of those CDL pixels over three study years (2006, 2012, 2014) with nearby reference points. Only the CDL core pixels that display the same land cover with the reference point and remain unchanged over the three study years were selected for training and validation as it is highly likely that LCLU types of those pixels from the CDL are correct. Despite known issues with the CDL as discussed above, we used it to enhance the reference dataset for three reasons. First, the major crops are mapped with very high accuracy. Second, by collapsing CDL classes to create broader categories and

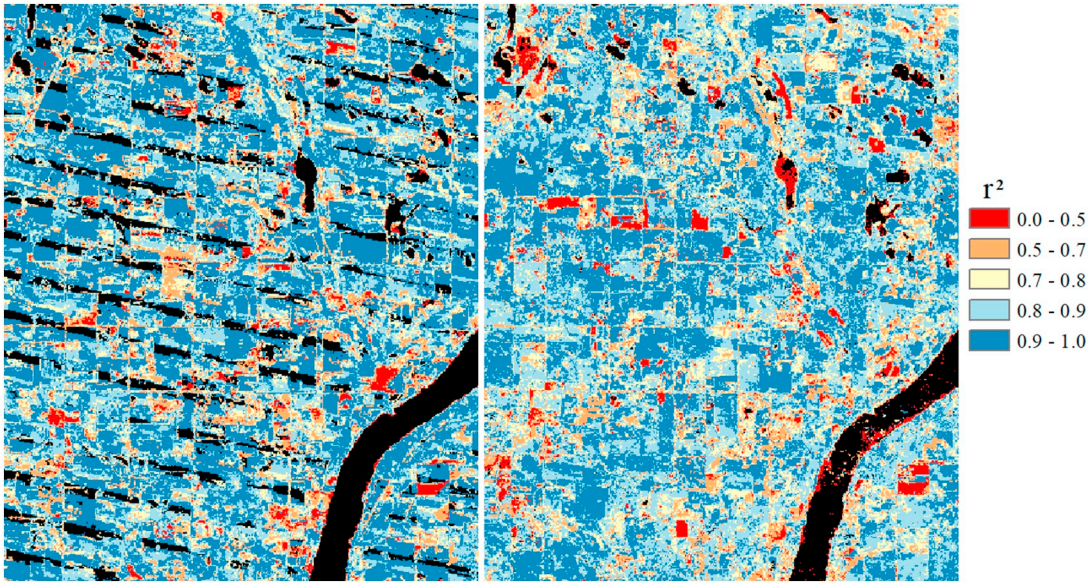


Fig. 3. Coefficients of determination (r^2) of the CxQ models for a sample area in Roberts County in 2012 (left) and 2014 (right). Black pixels indicate that no CxQ model was fitted due to a lack of valid observations. There were almost no observations from Landsat 5 in 2012 (end of operation), and the Scan Line Corrector (SLC) issue of Landsat 7 causes missing data generating the diagonal stripes.

Table 1
Fitted parameter coefficients, derived phenometrics, and other metrics from the CxQ model.

Parameters	Meaning
α, β, γ	Fitted parameter coefficients of CxQ model (Eq. (1))
TTP	Thermal time to peak (AGDD at the max fitted EVI) ($TTP = -\beta/2 \times \gamma$)
PH	Peak height EVI (max fitted EVI) ($PH = \alpha - \beta^2/4 \times \gamma$)
HTV	Value of EVI at half-TTP ($HTV = \alpha + \beta \times TTP/2 + \gamma \times TTP^2/4$)
ymax	Highest observed EVIs
r^2	Coefficient of determination of the fitted model
lpos, rpos	Observation index of start and end of the fitting window
o_fit	Number of observations used to fit the CxQ model
o_per	Ratio of "o_fit" to the total number of observations
minx, maxx	AGDD at left and right ends of the fitted curve in the first quadrant
peaks	Number of high EVI values ($\geq 0.8 * ymax$) outside the fitting window
jumps	Number of times that $\Delta EVI \geq 0.2$

searching for consistent land cover core patches over years, the chance to select "good" pixels (those that show true land cover information) for the sample datasets is high. Third, the CDL is the only publicly available LCLU cropland dataset with fine spatiotemporal resolution.

3.3.2. Classification and accuracy assessment

The RFC (Breiman, 2001) is an ensemble classification algorithm that constructs a set of decision trees to make a prediction. Each tree is created using a randomly selected subset of training samples and variables (Nvar). By growing the forest up to a user-defined number of trees (Ntree), the RFC creates a set of trees with high variance but low bias. The final classification result is generated by averaging the class assignment probabilities calculated across all produced trees. New unlabeled data inputs are evaluated against all decision trees created in the ensemble and each tree votes for a class membership. The membership class receiving maximum votes is selected. Conventional accuracy metrics (overall accuracy-OA, user accuracy-UA, producer accuracy – PA, and kappa statistics) were used to evaluate model performance. Each RFC run was performed using default values of Nvar (the square root of the number of predictor variables = $\sqrt{16} = 4$) and Ntree (500) using 'randomForest' package in R program (Liaw and Wiener, 2002).

3.3.3. Scenarios for constructing the sample dataset

To examine the sensitivity of the RFC to sample size and design, we generated sample datasets in different scenarios to use in the classification process. For each scenario, we generated 20 RFC runs in which 2/3 of the sample data were randomly selected for the training phase and the remainder reserved for the validation step.

(a) Increasing sample size with no control on sampling design

We first want to examine how RFC performed without any control on sampling design. This scenario served as a baseline to evaluate whether or not putting restrictions on the sampling design would help improve the RFC model performance. In this scenario, we selected CDL core pixels at various distances from the reference points, from 300 m to 1500 m at 100 m increments. All selected pixels within each buffer were used in the classification. Table 2 shows the size of sample datasets at 1500 m buffer in the initial list of study areas. To enable consistent comparisons of model performance, we excluded four counties (italics in Table 2) that have no data for training/validation in at least one land cover category.

(b) Increasing sample size with control on sampling design

RFC has been found to perform better with larger sample datasets (Deng and Wu, 2013; Du et al., 2015). The tradeoffs for better performance are higher costs in data collection and longer times in computation. A previous study suggested that the sample dataset should represent about 0.25% of the total study area for accurate classification using RFC (Colditz, 2015). RFC is also sensitive to the design of the sample dataset. In this scenario, we built a sample dataset from the 1500 m data pool so that it followed one of two options: either (1) the proportional distribution among classes in the sample dataset was equivalent to the proportional distribution found in the re-classified CDL layer ("Same Distribution") or (2) each class in the sample dataset contained the same number of pixels ("Same Size"). Both "Same Distribution" and "Same Size" sample datasets were generated to represent from 0.15% to 0.35% of the total county area (in 0.05% increments) of the total county area. Table 3 presents the maximum size of sample datasets constructed from 1500 m data pool for the two options as a percentage of the corresponding county area. Five counties highlighted

Table 2
Number of pixels in each land cover group at 1500 m buffer. Counties with zero pixels in at least one land cover group are highlighted in italics.

County. year	Cropland	Grassland	Others
Roberts.06	42,447	11,552	19,526
Roberts.12	38,218	11,461	17,547
Roberts.14	42,707	11,556	18,601
Codington.06	42,471	6448	9789
Codington.12	37,237	6425	7903
Codington.14	42,508	6445	9355
Kingsbury.06	99,221	11,905	8409
Kingsbury.12	89,253	11,892	2448
Kingsbury.14	100,400	11,901	1622
Miner.06	37,458	9224	1289
Miner.12	32,681	9165	1218
Miner.14	37,698	9232	1304
Hutchinson.06	154,241	16,586	1562
Hutchinson.12	150,543	16,289	1478
Hutchinson.14	155,510	16,233	1541
Bon_Homme.06	46,719	2538	13,220
Bon_Homme.12	44,574	2529	9715
Bon_Homme.14	47,176	2521	12,875
Buffalo.06	8121	195,519	10
Buffalo.12	7522	196,922	12
Buffalo.14	7943	199,560	16
Campbell.06	4954	59,724	5397
Campbell.12	3519	78,272	19
Campbell.14	4975	78,342	1138
Walworth.06	13,580	107,057	5572
Walworth.12	10,292	115,960	2284
Walworth.14	13,613	115,711	2959
Hand.06	82,887	150,825	0
Hand.12	77,461	150,601	0
Hand.14	83,445	150,228	0
Hanson.06	51,212	4695	0
Hanson.12	45,742	4695	0
Hanson.14	51,413	4689	0
Jones.06	1417	469,531	0
Jones.12	1405	475,311	0
Jones.14	1416	477,774	0
Lawrence.06	0	0	356,281
Lawrence.12	0	0	356,979
Lawrence.14	0	0	356,851

in italics had sample datasets covering < 0.35% of their county areas in at least one of three years. To compare these sample designs, we examined only the four counties that had sufficient sample pixels support both designs: Bon Homme, Codington, Roberts, and Walworth.

3.3.4. Major crops mapping

To understand classification errors of alternative RFC models (Section 3.3.3) and to examine the capability of LSP-based classification in characterizing major crops, we conducted land cover mapping for Codington and Roberts in 2012 and 2014 because they have sufficient sample data with good quality derived from the CDL. Three broad land use classes from the previous processing were divided into finer classes. “Cropland” had four classes: corn, soybean, wheat, and other crops. “Others” was separated into water, developed, and other natural (mostly contains forest and wetland). “Grassland” remained as grassland.

3.4. Cross-comparison between RFC model outputs and the Cropland Data Layer

We compared estimated areas from RFC models with those from reclassified CDL, in addition to using conventional accuracy metrics to evaluate the performance of classifications. It is important to note that the CxQ modeling may fail due to a lack of valid Landsat observations yielding “no-data” (Section 3.2, Fig. 3). These “no-data” pixels were not classified in RFC models and generated gaps in the output. We masked the reclassified CDL by “valid pixels” from RFC to allow cross-comparison between the two datasets. Due to variation in quantity and quality of Landsat data over time, the RFC outputs contain gaps in different places across the years. Therefore, direct comparison of land cover area between annual RFC outputs is not appropriate and do not indicate actual changes in the study area. However, since estimation from RFC and CDL each year are based on the same ‘valid pixels’ raster, we can compare changes in land cover areas as estimated by RFC and by the re-classified CDL. This indirect measurement gave us an idea about how well RFC models performed.

Table 3
Maximum size of sample datasets constructed from 1500 m buffer data pool (Table 2) as a percentage to county area. Counties with the sample size covering < 0.34% of their areas for at least one year are highlighted in *italics*.

County	Same Distribution		
	2006	2012	2014
Roberts	0.85	1.39	1.36
Codington	1.06	1.04	1.12
Kingsbury	1.76	1.20	0.81
Miner	0.51	0.95	1.05
Hutchinson	0.52	0.87	0.92
Bon Homme	0.64	0.57	0.58
Buffalo	0.01	0.03	0.03
Campbell	0.82	0.02	0.15
Walworth	1.65	0.34	0.36

County	Same Size		
	2006	2012	2014
Roberts	1.10	1.17	1.11
Codington	1.00	1.10	1.02
Kingsbury	1.09	0.34	0.21
Miner	0.24	0.24	0.24
Hutchinson	0.20	0.20	0.20
Bon Homme	0.48	0.49	0.47
Buffalo	0.00	0.00	0.00
Campbell	0.75	0.00	0.16
Walworth	0.83	0.37	0.44

4. Results

4.1. Increasing sample size with no control on designs

We examined how the RFC responses to different sample sizes created by collecting pixels within 300 m to 1500 m from reference points. Table 4 contains the accuracy metrics means, estimated from running the RFC 260 times (20 trials × 13 sample scenarios) for each county-year. Accuracy metrics indicate good performance of the RFC for the four counties in 2006, 2012, and 2014. Performance in 2012 is slightly weaker than the other two years, possibly due to drier, warmer weather and early crop planting in eastern SD in 2012. Standard deviation values are always < 5% of mean values for all cases indicating that different sample sizes did not significantly affect RFC accuracy (Table 4).

Fig. 4 presents predicted areas (mean and standard deviation from 20 runs) for each land cover type by the RFC based on sample datasets collected within 300 m to 1500 m (in 100-m increments) of reference points. Although accuracy metrics of those models are all quite good with only small variations, their predicted values can be very different from each other and from the CDL value, especially for the “grassland” and “others” categories. Among the three classes, predicted cropland areas are the most consistent among different sample scenarios and they are closest to CDL values. There is no surprise that cropland is the best predicted class due to more pronounced phenological characteristics and the higher accuracy of crops in the CDL. The “others” class is the least accurate group; again, not surprising as there are many non-vegetated surfaces included in this heterogeneous residual category.

Table 4
Accuracy assessment of the RFC models with no control on design. OA is overall accuracy; PA is producer's accuracy; UA(σ) is user's accuracy and its standard deviation in percent to mean value; and kappa is Cohen's kappa.

Year	Kappa	OA(σ)	Cropland		Grassland		Others	
			PA	UA	PA	UA	PA	UA
(a) Bon Homme								
2006	0.96	0.98 (0.08%)	0.98	0.99	0.91	0.95	0.99	0.96
2012	0.87	0.94 (0.64%)	0.93	0.98	0.93	0.91	0.96	0.84
2014	0.98	0.99 (0.07%)	0.99	0.99	0.93	0.95	0.99	0.98
(b) Codington								
2006	0.98	0.99 (0.06%)	0.99	0.99	0.96	0.94	0.98	0.98
2012	0.81	0.90 (1.08%)	0.86	0.99	0.89	0.87	0.97	0.77
2014	0.97	0.98 (0.37%)	0.99	0.99	0.92	0.88	0.97	0.98
(c) Roberts								
2006	0.96	0.97 (0.18%)	0.98	0.98	0.97	0.96	0.97	0.96
2012	0.87	0.92 (0.59%)	0.88	0.99	0.94	0.94	0.97	0.83
2014	0.95	0.97 (0.26%)	0.98	0.99	0.95	0.92	0.95	0.96
(d) Walworth								
2006	0.93	0.96 (0.39%)	0.95	0.98	0.98	0.96	0.90	0.96
2012	0.89	0.94 (1.13%)	0.76	0.99	0.99	0.97	0.92	0.82
2014	0.96	0.98 (0.18%)	0.99	0.99	0.99	0.97	0.91	0.98

Fig. 4 also indicates that larger sample size, in some cases, does not lead to better prediction: predicted values departed from the baseline at longer buffer distances. Across all trials, the RFC generally overestimated land cover class with the largest area.

4.2. "Same distribution" vs. "same size" models

Table 5 contains the RFC models accuracy metrics using "Same Distribution" and "Same Size" sample datasets (mean values calculated from 5 × 20 = 100 models). The high accuracy values indicate that the RFC performed well in almost every case, except producer's accuracies of "others" class for 2012 and 2014 in Walworth County (bold in Table 5). Both producer's and user's accuracies of cropland are consistently higher than the other two classes in all years and every county. The "Same Distribution" RFC tends to perform better than 'Same Size' models, as shown in higher overall accuracy as well as kappa scores.

Fig. 5 provides the RFC performance of "Same Distribution" and "Same Size" sample scenarios for Codington. In both cases, these RFC models seemed to have better cropland and grassland area estimates than the models from Fig. 4 (*i.e.*, increasing sample size due to larger buffer and no control on sample designs) as predicted values of control scenarios are located closer to the baseline value from CDL. Overall, "Same Size" models tend to underestimate cropland area compared to "Same Distribution" models. However, there is no clear improvement between "Same Distribution" and "Same Size" in cropland area estimation, as one scenario can be better in some cases but worse in the others (Figs. 5, S1–S3). However, "Same Distribution" models clearly perform better for the other two groups: the "Same Distribution" values (squares) tend to locate closer to the CDL baseline than the "Same Size" values (circles).

4.3. Influence of variable importance on model performance

Published studies have indicated that although the RFC can work with a large amount of input variable, high dimensional input may weaken performance of the RFC (Millard and Richardson, 2013, 2015). To better understand the output of previous analyses and to improve the RFC performance, we evaluated variable importance from 720 RFC models (4 counties × 3 years × 3 sample sizes × 20 runs) in the 'Same Distribution' models (since these had performed better in earlier evaluation scenarios) in seven groups including four by-county groups (Roberts, Codington, Walworth, and Bon Homme) and three by-year groups (2006, 2012, 2014). In each group, we calculated the sum of "Mean Decrease Gini" to evaluate variable importance. To treat variable importance in each group equally, we graded each variable by its "Mean Decrease in Gini" from 16 to 1 in descending order of importance. A higher summation of grade over the seven groups (rescaled to 0–1) indicated a more overall important variable. From the seven groups, we also counted a number of times each variable placed in the top eight ranked important variables. The top eight important variables appear in Table 6. It is no surprise that the three parameter coefficients of the quadratic curve are three most important variables as shape of the fitted curve is defined by these coefficients. Other phenometrics were calculated from the fitted values of α, β, γ, including TTP, PH, HTV, xmin, xmax. Histograms of α, β, γ fitted values for each county-year are presented in Tables S8–S11.

We compared performance of the RFC with all 16 variables ('full') and the top eight important variables ('reduced') in estimation of cropland area for all study counties in three years due to much higher and consistent accuracy of cropland prediction over the other two land cover type. Fig. 6 indicates that the 'reduced variable' models might perform slightly better than the 'full variable' models, but differences between the two are not striking. In 2006 and 2012, cropland areas seem to be underestimated by the reduced models compared to the full one: unfilled circles tend to fall closer to the dashed line (Fig. 6). In 2014, estimations from both sets of models were very similar. Sample land cover maps produced by "reduced variables" models are provided in Figs. S12–S15.

Considering that the reduced variable models were more likely to return the better estimation of cropland area, we used the output of those models to compare changes of cropland areas as predicted by our study with the re-classified CDL layers. Tables 7 & 8 present estimated cropland for the full spatiotemporal coverage by RFC and CDL as well as differences in cropland areas between different periods. Cropland area estimates were similar for the 'reduced variable' and the 'full variable' models. The changes in cropland areas were very different between RFC and CDL for the 2006–2012 period, possibly due to (1) lower accuracy of the 2006, 2012 CDLs than the 2014 CDL (USDA-NASS 2018), and (2) the crop failures due to widespread drought in 2012. The changes in cropland areas were more similar for the 2012–2014 period (Table 8). In the 2012–2014, the CDL tends to underestimate the increase of cropland compared to RFC estimation possibly due to area of crop failure areas in 2012 detected by the RFC as "grassland" or "others".

4.4. Mapping of major crops

We performed land cover classification with more detailed cover types (corn, soybean, wheat, other crops, grassland, other natural, developed, and water) for Roberts and Codington counties. Tables 9 and 10 show accuracy metrics and estimated land cover areas of RFC models. Overall accuracies for all four scenarios are > 0.75. The 2014 models performed much better than the 2012 models, not a surprise considering that the 2014 CDL has higher accuracy than the 2012 CDL. Among the land cover types, "water" has the highest user's/producer's accuracy despite the poor CxQ fit, since the EVI time series of water pixels always nearly flat. On the other hand, "other natural" and "developed" were classified with lower accuracy due to their complex phenological characteristics. While "other natural" consists of minor vegetation classes with quite different LSP pattern, *e.g.*, forest and wetland, "developed" encompasses by heterogeneous surfaces and materials, including both vegetated and non-vegetated surfaces. Estimated land cover areas are most accurate for corn and soybean due to their distinctive phenological characteristics and high CxQ fits. Many vegetated pixels within the urban boundary are identified as either grassland or other vegetation leading to underestimation of the urban area. On the other hand, RFC models overestimated grassland area because other land cover types could be misclassified as "grassland",

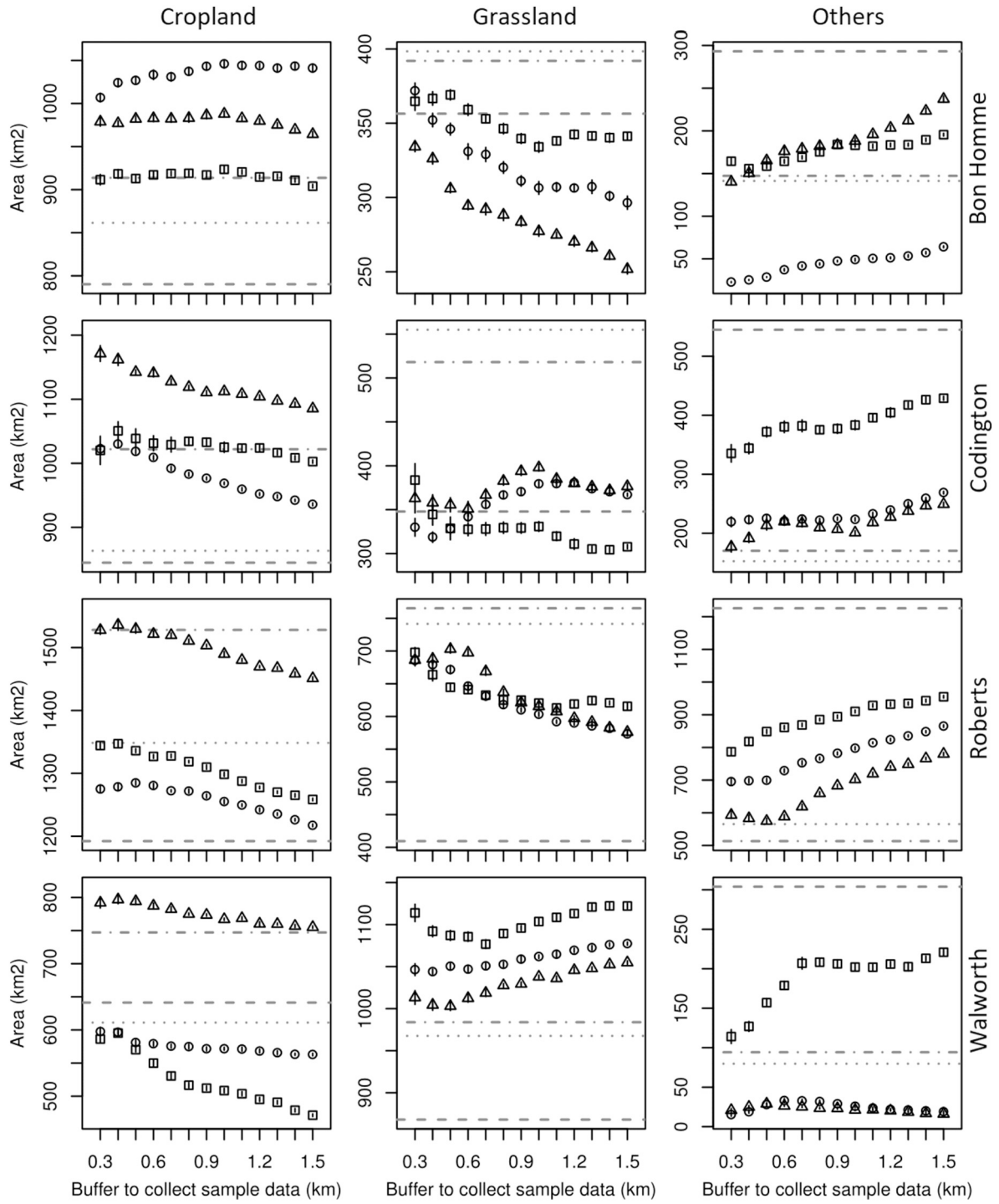


Fig. 4. Predicted areas for each land cover type (mean ± 1σ) by the RFC in 2006 (square), 2012 (circle), and 2014 (triangle). Dashed, dotted, and dot-dash lines present estimated areas using re-classified 2006, 2012, and 2014 CDL, respectively, as baselines for evaluation.

e.g., green spaces within the urban area, failed crop pixels or dry wetland pixels.

5. Discussion

5.1. Sensitivity of random forest classification to sample size, design and quality

Our findings confirm previous findings that RFC is sensitive to sample size (Deng and Wu, 2013; Du et al., 2015; Millard and Richardson, 2015). However, while previous studies indicated that larger sample size increases RFC accuracy, our results show that is not always the case. Sometimes, larger sample dataset may reduce RFC accuracy depending on the quality of the sample dataset. Our study showed that a sample dataset covering 0.15%–0.35% of the study area is adequate to achieve an accurate land cover classification, similar to the findings of *Colditz* (2015).

As published studies suggest that the proportional distribution of classes in the sample dataset should represent the proportional distribution of those classes in the target population (Dalponte et al., 2013; Millard and Richardson, 2013, 2015), we tested RFC performance with different sample designs. Our results also showed that sample dataset

Table 5
Accuracy assessment of RFC for "Same Distribution" and "Same Size" models. OA is overall accuracy; PA is producer's accuracy; UA is user's accuracy; and kappa is Cohen's kappa. Accuracy metrics of < 0.7 are highlighted in **bold**.

Year	Same Distribution								Same Size							
	Kappa	OA	Cropland		Grassland		Others		Kappa	OA	Cropland		Grassland		Others	
			PA	UA	PA	UA	PA	UA			PA	UA	PA	UA	PA	UA
(a) Bon Homme																
2006	0.95	0.97	0.98	0.99	0.95	0.95	0.93	0.93	0.94	0.96	0.97	0.98	0.95	0.95	0.95	0.94
2012	0.90	0.94	0.96	0.95	0.97	0.93	0.78	0.93	0.90	0.93	0.90	0.92	0.98	0.94	0.91	0.94
2014	0.96	0.97	0.99	0.99	0.97	0.96	0.90	0.92	0.94	0.96	0.98	0.99	0.96	0.95	0.95	0.95
(b) Codington																
2006	0.95	0.97	0.98	0.99	0.94	0.94	0.95	0.95	0.95	0.96	0.98	0.99	0.96	0.95	0.95	0.95
2012	0.91	0.95	0.98	0.99	0.96	0.92	0.77	0.85	0.88	0.92	0.96	0.98	0.91	0.88	0.89	0.90
2014	0.93	0.96	0.99	0.99	0.96	0.92	0.77	0.89	0.88	0.92	0.98	0.99	0.92	0.86	0.86	0.92
(c) Roberts																
2006	0.91	0.95	0.95	0.96	0.92	0.92	0.94	0.94	0.92	0.95	0.95	0.96	0.97	0.95	0.92	0.93
2012	0.91	0.94	0.98	0.97	0.94	0.92	0.85	0.89	0.89	0.93	0.96	0.96	0.93	0.92	0.89	0.90
2014	0.91	0.94	0.97	0.98	0.96	0.92	0.84	0.87	0.89	0.93	0.95	0.98	0.94	0.90	0.89	0.90
(d) Walworth																
2006	0.87	0.92	0.95	0.95	0.94	0.93	0.78	0.82	0.86	0.90	0.94	0.95	0.90	0.89	0.88	0.87
2012	0.90	0.95	0.99	0.99	0.99	0.93	**0.23**	0.78	0.82	0.88	0.98	0.99	0.82	0.83	0.84	0.83
2014	0.92	0.96	0.99	0.99	0.98	0.94	**0.40**	0.84	0.86	0.90	0.99	0.99	0.85	0.86	0.86	0.86

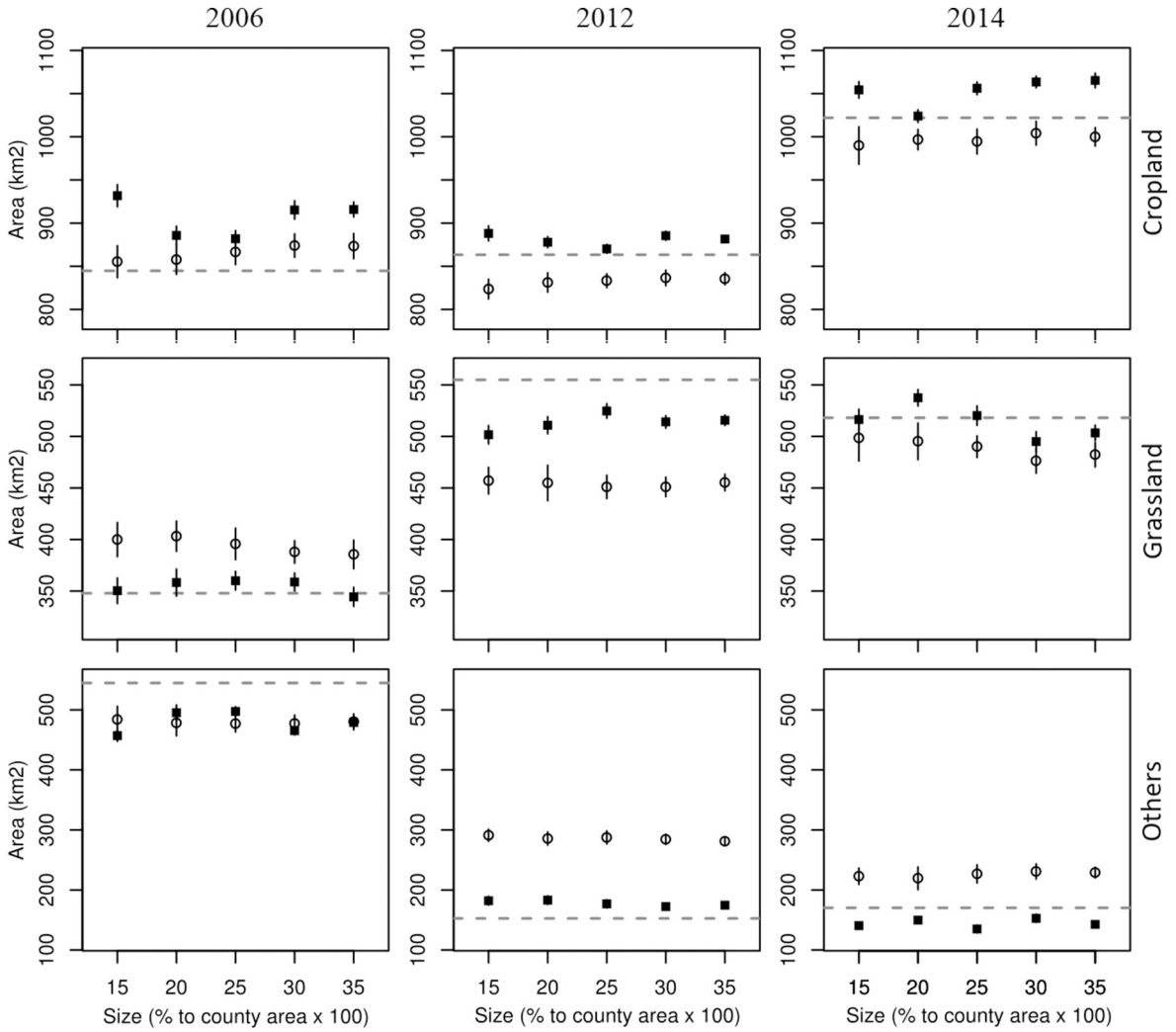


Fig. 5. Area of each land cover type (mean ± 1σ) estimated by 'Same Distribution' (square) and 'Same Size' (circle) RFC models for Codington. Dashed lines present areas of corresponding land cover types estimated by re-classified CDL.

Table 6
Top eight important variables for the random forest classification.

Variables	Meaning	Count (max of 7)	Grade (range: 0–1)
α	Constant component of the CxQ model	7	0.92
β	Linear component of the CxQ model	7	0.91
γ	Quadratic component of the CxQ curve	7	0.82
o_per	Ratio of 'o_fit' to the total number of observations	5	0.49
minx	AGDD at the left end of the fitted curve	5	0.47
ttp	AGDD at the max fitted EVI	5	0.46
ph	Max fitted EVI	4	0.37
o_fit	Number of observations used to fit the CxQ model	4	0.34

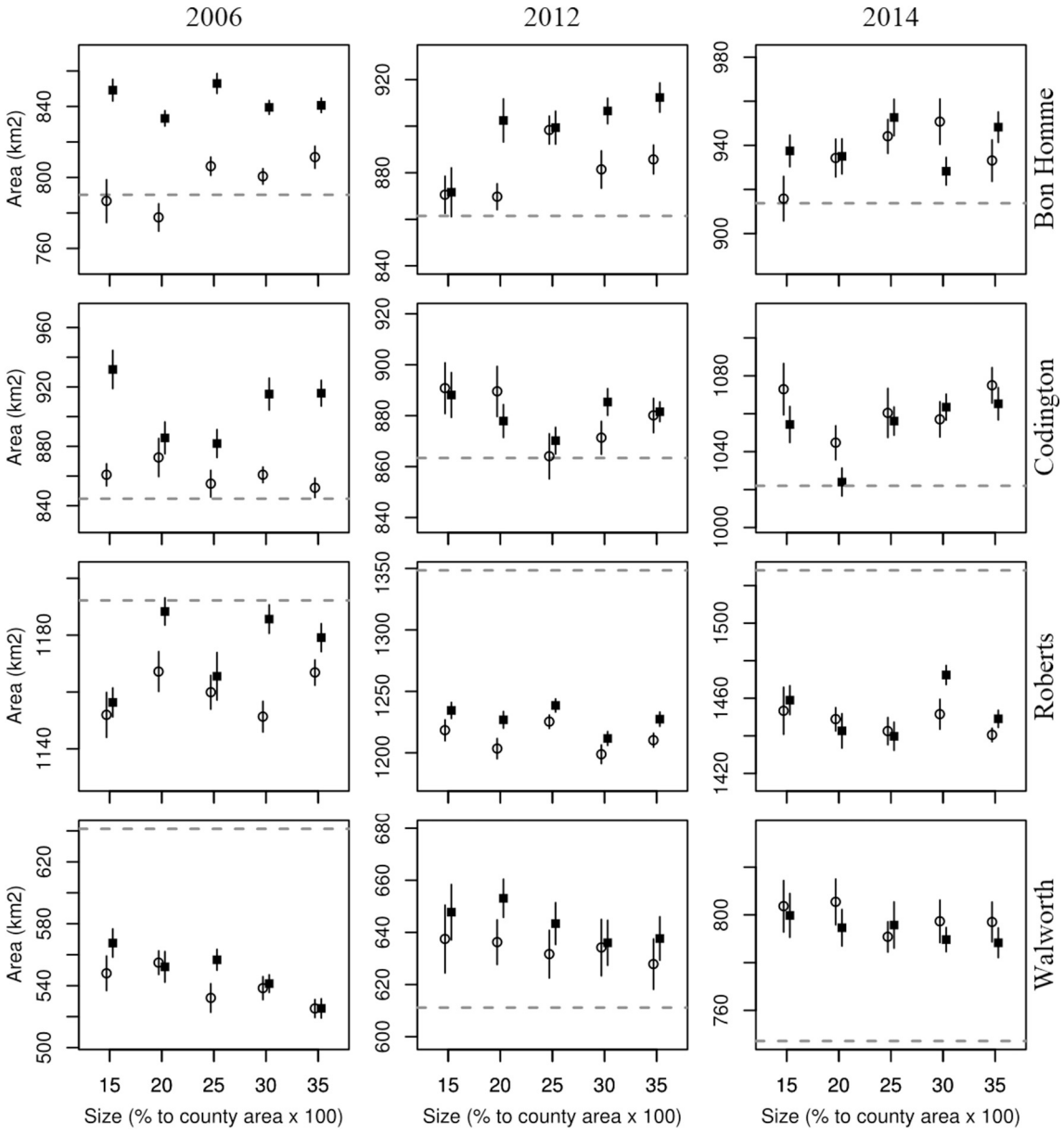


Fig. 6. Comparison between cropland areas (mean ± 1σ) estimated by "full variable" (square) and "reduced variable" (circle) RFC models. Dashed lines present cropland areas estimated by re-classified CDL.

created by controlling the proportional distribution of data in each land cover group ("Same Distribution") yielded better RFC performance than those with no control ("Same Size"). However, we did not find substantial differences between 'Same Distribution' and 'Same Size' models, possibly because the performance of the "Same Size" models were already quite good.

We leveraged the CDL dataset to augment the number of training samples; thus, our results strongly depend on the accuracy of the CDL layers. Although we sought pixels with correct land cover information by using reference data, avoiding edge pixels and matching information for multiple years, there is a good chance that the training sample dataset still contains a considerable but unquantifiable amount of

Table 7
Comparison between cropland areas (in km^2) estimated by the "full variable" and the "reduced variable" RFC models and those from re-classified CDL for the four study counties.

	2006		2012		2014	
	Full	Reduced	Full	Reduced	Full	Reduced
BonHomme_RFC	843	796	898	881	940	936
BonHomme_CDL	790	790	861	861	914	914
Diff w/CDL (%)	6.7%	0.8%	4.3%	2.3%	2.9%	2.4%
Codington_RFC	906	860	881	879	1053	1062
Codington_CDL	845	845	863	863	1022	1022
Diff w/CDL (%)	7.3%	1.8%	2.0%	1.8%	3.0%	3.9%
Roberts_RFC	1175	1159	1228	1211	1453	1447
Roberts_CDL	1192	1192	1348	1348	1528	1528
Diff w/CDL (%)	−1.5%	−2.8%	−9.0%	−10.2%	−4.9%	−5.3%
Walworth_RFC	549	540	644	633	794	799
Walworth_CDL	641	641	611	611	474	747
Diff w/CDL (%)	−14.5%	−15.8%	5.31	3.7	6.2%	6.9%

Table 8
Changes in cropland (in km^2) estimated by the "full variable" and "reduced variable" RFC models and percent difference from changes in cropland estimated by the re-classified CDL.

	Δ2006–2012		Δ2012–2014	
	Full	Reduced	Full	Reduced
BonHomme_RFC	55.33	84.63	41.86	54.47
BonHomme_CDL	71.23	71.23	52.30	52.30
Diff w/CDL(%)	−22.3%	18.8%	−20.0%	4.2%
Codington_RFC	−25.45	18.93	172.00	182.86
Codington_CDL	18.60	18.60	158.60	158.60
Diff w/CDL (%)	−236.8%	1.8%	8.5%	15.3%
Roberts_RFC	52.80	51.79	224.79	236.11
Roberts_CDL	156.19	156.19	179.66	179.66
Diff w/CDL (%)	−66.2%	−66.8%	25.1%	31.4%
Walworth_RFC	94.95	93.74	150.04	165.35
Walworth_CDL	−30.16	−30.16	136.01	136.01
Diff w/CDL (%)	−414.8%	−410.8%	10.3%	21.6%

Table 9
Accuracy assessment of RFC models (mean of 60 models with sample dataset covering 0.15%, 0.25%, 0.35% of the total area) for Codington and Roberts counties. PA is producer's accuracy; UA is user's accuracy; kappa is Cohen's kappa.

	Codington 2012	Codington 2014	Roberts 2012	Roberts 2014
Overall Accuracy	0.763	0.871	0.796	0.844
kappa	0.696	0.837	0.743	0.804

	Codington 2012		Codington 2014		Roberts 2012		Roberts 2014	
	PA	UA	PA	UA	PA	UA	PA	UA
Corn	0.80	0.77	0.90	0.89	0.86	0.80	0.90	0.89
Soybean	0.66	0.71	0.89	0.91	0.80	0.89	0.89	0.91
Wheat	0.69	0.76	0.93	0.90	0.74	0.81	0.92	0.88
Other Crops	0.62	0.78	0.77	0.82	0.57	0.82	0.67	0.87
Water	0.89	0.88	0.95	0.96	0.93	0.92	0.96	0.96
Developed	0.29	0.51	0.43	0.68	0.27	0.47	0.30	0.51
Other Natural	0.30	0.63	0.28	0.61	0.66	0.63	0.66	0.66
Grassland	0.92	0.79	0.93	0.84	0.89	0.81	0.91	0.83

classification error. The overall accuracy of crop classes (the most accurate classes) in South Dakota are only 61.2% in 2006, 74.7% in 2012 and 86.8% in 2014 (USDA-NASS, 2018). The 2006 and 2012 CDLs have lower accuracies than the 2014 CDL, possibly due to the start of a new CDL classification system in 2006 (Boryan et al., 2011), and the unusually early spring, subsequent drought (Ault et al., 2013) as well as the lack of observations due to the end of Landsat 5 operation in 2012.

5.2. Accuracy metrics and prediction consistency of RFC

Generally, land cover/land use studies must work with limited samples due to the high cost of conducting field observations. Thus, most attention focuses on retrieving an accurate classification model using a single sample dataset. The RFC performance depends on the actual sample dataset in hand and how it is used to train the RFC. So, an important question is how the model accuracy and predicted land cover map might change were we to have the opportunity to collect the second sample dataset independent of the first. By performing multiple RFC models on different sample sizes and scenarios, we were able to test something that has received relatively little attention in the literature (but see Jin et al., 2014; Colditz, 2015; Millard and Richardson, 2015; Lyons et al., 2018). Our results show that even if the RFC models have quite good accuracy, their predicted values still can be quite different from each other and from the real population. This pattern of performance suggests that an ensemble RFC output generated over multiple scenarios is important, even if there is just one training sample dataset available.

Here, we have assumed that RFC models with outputs closer to the re-classified CDL are better. However, this assumption may not be true due to low accuracies in the CDL, even in some crop classes. Despite the differences between our estimates with those from CDL, the performance of RFC in our study is quite consistent through multiple scenarios and runs. It means that predicted land cover maps derived from different sample selections would be very similar to each other, and the only option to get a better model would be to improve the quality of sample data.

5.3. Weaknesses and prospects for the proposed classification approach

Due to variation in data quality, EVI time series at different locations are often not observed at the same set of days, preventing a direct comparison between annual EVI patterns. Here, we distilled the entire annual AGDD and EVI time series into a simple functional form so that land surface phenology at every pixel was described by the same set of parameters (though different fitted parameter coefficients). The attempt to use a single function for every land cover was only partially successful. The CxQ models fitted well most vegetated surfaces, especially for cropland pixels. Median r^2 in Roberts County were approximately 0.77 in 2006, and 0.90 in 2012 and 2014 (Fig. S16). In contrast, non-vegetated pixels had low or sometimes negative EVI leading to very poor model fits or even model failure that generates gaps in the spatial coverage. The quality of the model fits was not consistent between land cover types: higher quality for crops, medium quality for non-crop vegetation including grassland, and lower quality for non-vegetated surfaces (Fig. 3). The quality of model fits also varied within a class due to seasonal and interannual climatic variation as well as gaps resulting from the Scan Line Corrector failure on Landsat 7 ETM+. These inconsistencies in CxQ LSP model fitting reduces the strength of using just the LSP coefficients and phenometrics to map land cover and track changes. Another limitation of LSP-based approach was the gap creation during the fitting process arising from model failure due to low number of good Landsat observations in some years and over some areas (*cf.* Section 3.2). Because the gaps were produced in different places across the years, a direct comparison between yearly RFC outputs was not appropriate in most cases, without spatio-temporal interpolation, something that we chose not to do. Lastly, except for croplands with very distinctive and consistent phenological shapes for the CxQ model to capture, much non-crop vegetation can exhibit similar phenological curves and phenometrics, making it difficult to map that those land cover types accurately. For example, CxQ curves fitted

Table 10
Land cover areas (in km^2) estimated by RFC and by reclassified CDL and percent difference in area estimated for Codington (C) and Roberts (R) counties.

	Corn	Soybean	Wheat	Other Crops	Water	Developed	Other Natural	Grassland	Total Area
C.RFC.2012	417	296	59	44	50	51	17	637	1571
C.CDL.2012	419	307	76	63	17	98	36	555	1571
%Diff	−0.3	−3.6	−21.8	−29.6	200.7	−48.2	−54.3	14.8	–
C.RFC.2014	454	366	101	66	39	54	12	620	1721
C.CDL.2014	419	430	91	82	39	103	27	518	1709
%Diff	8.2	−15.0	11.1	−19.9	−1.0	−48.0	−56.1	19.8	–
R.RFC.2012	613	504	68	17	43	65	510	836	2656
R.CDL.2012	637	597	79	36	40	126	398	741	2654
%Diff	−3.7	−15.6	−14.3	−54.3	7.4	−48.3	28.0	12.8	–
R.RFC.2014	624	688	96	29	42	83	424	820	2806
R.CDL.2014	604	788	90	50	51	129	330	765	2807
%Diff	3.3	−12.7	6.5	−41.0	−16.5	−35.6	28.7	7.1	–

for grasslands can appear like ones fitted to wetlands during drier weather. Many lawns or recreational fields in urban areas can be misclassified as grassland, which is not completely wrong, but requires contextual information to be classified more accurately as “developed”.

Despite all limitations as discussed above, the future of LSP-based classification is promising for several reasons. First, although LSP-based classification may not work well for non-cropped surfaces, this method may still be useful for detailed crop mapping. Our sample analyses demonstrated that we could discriminate quite well between three major commodity crop types: corn, soybean, and wheat. Second, here we used only LSP-based variables as input to the RFC; however, our approach could be expanded to take advantage of additional time series from other spectral bands and specific indices tailored to reveal different aspects of the dynamic land surface. Third, although the LSP-fitting process generated gaps in the output, bringing together complementary sensor datastreams, such as from Sentinel 2A and 2B, could increase the temporal density of observations and thereby increase the quality of the time series to be modeled. In areas with persistent cloud cover, it may be possible to leverage “CubeSats” with datastreams at even higher spatial and temporal but lower spectral resolution to infill gaps intelligently using physical-based modeling (Houborg and McCabe, 2018).

6. Conclusions

We have introduced a new approach to characterize land cover/land use using Landsat image time series, *first*, by modeling the seasonality of the vegetated land surface at the pixel scale using Landsat and MODIS time series linked with a simple parametric model of land surface phenology and, *second*, by submitting the per-pixel fitted model coefficients and associated phenometrics to a random forest classifier (RFC). The LSP-based classification was demonstrated through a comparative analysis on four counties in South Dakota over three separate years. Our results showed that the classification based only on LSP could accurately differentiate major commodity crops, but it had limited accuracy for non-vegetated classes. To evaluate the differential performance of RFC models arising from the training sample dataset, we created several distinct training sample datasets from reference points and CDL layers using different sample sizes and designs. Among all sampling designs, the “Same Distribution” models tended to yield marginally better predictions of land cover. In addition, the RFC models used only the most eight important variables (*i.e.*, the three fitted parameter coefficients (α, β, γ); the maximum model EVI; the AGDD at the maximum modeled EVI; the number of observations used to fit CxQ model; and the number of valid observations) had slightly higher accuracy compared to those using all variables. Although all the RFC models achieved quite good accuracy metrics, there were substantial differences in predicted land cover areas between alternative RFC models. As each sample dataset will yield different prediction, we suggest that future supervised classification studies should carefully examine output stability. Ideally, the final prediction should be an ensemble of all outputs retrieved by performing classification with multiple independent samples. However, even if only one sample dataset is available, the final output will still benefit from iterating training/validation and prediction multiple times.

Although the current results show limitations of the LSP-based classification for non-vegetated surfaces, the prospect for using land surface phenology variables in land cover/land use classification is promising. First, the LSP-based classification offered an accurate, consistent way to identify cropland areas and changes between major crops (*e.g.*, corn-soybean rotation) across years. Second, it should be possible to combine LSP variables with other indices and spectral information to improve classification accuracy and consistency, and advance the monitoring of land change using Landsat and comparable sensor datastreams and archives.

Acknowledgements

L.H. Nguyen and G.M. Henebry were supported, in part, by NASA project NNX14AJ32G Change in our MIDST: Detection and Analysis of Land Surface Dynamics in North and South America Using Multiple Sensor Datastreams, the Geospatial Sciences Center of Excellence at South Dakota State University, and the Center for Global Change and Earth Observations at Michigan State University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2018.12.016.

References

Ahl, D.E., Gower, S.T., Burrows, S.N., Shabanov, N.V., Myneni, R.B., Knyazikhin, Y., 2006. Monitoring spring canopy phenology of a deciduous broadleaf forest using MODIS. Remote Sens. Environ. 104 (1), 88–95.
Ault, T.R., Henebry, G.M., de Beurs, K.M., Schwartz, M.D., Betancourt, J.L., Moore, D., 2013. The false spring of 2012, earliest in North American record. EOS Trans. Am. Geophys. Union 94 (20), 181–182.
Baumann, M., Ozdogan, M., Richardson, A.D., Radeloff, V.C., 2017. Phenology from Landsat when data is scarce: using MODIS and dynamic time-warping to combine multi-year Landsat imagery to derive annual phenology curves. Int. J. Appl. Earth Obs. Geoinf. 54, 72–83.
Beck, P.S., Atzberger, C., Høgda, K.A., Johansen, B., Skidmore, A.K., 2006. Improved monitoring of vegetation dynamics at very high latitudes: a new method using MODIS NDVI. Remote Sens. Environ. 100 (3), 321–334.
de Beurs, K.M., Henebry, G.M., 2004. Land surface phenology, climatic variation, and institutional change: analyzing agricultural land cover change in Kazakhstan. Remote Sens. Environ. 89 (4), 497–509.
Boryan, C., Yang, Z., Mueller, R., Craig, M., 2011. Monitoring US agriculture: the US Department Of Agriculture, National Agricultural Statistics Service, Cropland Data Layer program. Geocarto Int. 26 (5), 341–358.
Breiman, L., 2001. Random forests. Mach. Learn. 45 (1), 5–32.
Claassen, R., Carriazo, F., Cooper, J.C., Hellerstein, D., Ueda, K., 2011. Grassland to cropland conversion in the Northern Plains. In: Economic Research Report. 120.
Clerici, N., Weissteiner, C.J., Gerard, F., 2012. Exploring the use of MODIS NDVI-based

phenology indicators for classifying forest general habitat categories. Remote Sens. 4 (6), 1781–1803.
Colditz, R.R., 2015. An evaluation of different training sample allocation schemes for discrete and continuous land cover classification using decision tree-based algorithms. Remote Sens. 7 (8), 9655–9681.
Dalponte, M., Orka, H.O., Gobakken, T., Gianelle, D., Næsset, E., 2013. Tree species classification in boreal forests with hyperspectral data. IEEE Trans. Geosci. Remote Sens. 51 (5), 2632–2645.
Deng, C., Wu, C., 2013. The use of single-date MODIS imagery for estimating large-scale urban impervious surface fraction with spectral mixture analysis and machine learning techniques. ISPRS J. Photogramm. Remote Sens. 86, 100–110.
Du, P., Samat, A., Waske, B., Liu, S., Li, Z., 2015. Random forest and rotation forest for fully polarized SAR image classification using polarimetric and spatial features. ISPRS J. Photogramm. Remote Sens. 105, 38–53.
Faber, S., Rundquist, S., Male, T., 2012. Plowed under: How Crop Subsidies Contribute to Massive Habitat Losses. Environmental Working Group, Washington, DC.
Fargione, J.E., Cooper, T.R., Flaspohler, D.J., Hill, J., Lehman, C., McCoy, T., ... Tilman, D., 2009. Bioenergy and wildlife: threats and opportunities for grassland conservation. Bioscience 59 (9), 767–777.
Franklin, S.E., Ahmed, O.S., Wulder, M.A., White, J.C., Hermosilla, T., Coops, N.C., 2015. Large area mapping of annual land cover dynamics using multi-temporal change detection and classification of Landsat time-series data. Can. J. Remote. Sens. 41, 293–314.
Friedl, M.A., Sulla-Menashe, D., Tan, B., Schneider, A., Ramankutty, N., Sibley, A., Huang, X., 2010. MODIS collection 5 global land cover: algorithm refinements and characterization of new datasets. Remote Sens. Environ. 114 (1), 168–182.
Gu, Y., Brown, J.F., Miura, T., Van Leeuwen, W.J., Reed, B.C., 2010. Phenological classification of the United States: a geographic framework for extending multi-sensor timeseries data. Remote Sens. 2 (2), 526–544.
Henebry, G.M., de Beurs, K.M., 2013. Remote sensing of land surface phenology: a prospectus. In: Schwartz, M.D. (Ed.), Phenology: An Integrative Environmental Science, 2e. Springer, pp. 385–411.
Houborg, R., McCabe, M.F., 2018. A cubesat enabled spatio-temporal enhancement method (CESTEM) utilizing Planet, Landsat and MODIS data. Remote Sens. Environ. 209, 211–226.
Huete, A., Didan, K., Miura, T., Rodriguez, E.P., Gao, X., Ferreira, L.G., 2002. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. Remote Sens. Environ. 83 (1–2), 195–213.
Jin, H., Stehman, S.V., Mountrakis, G., 2014. Assessing the impact of training sample selection on accuracy of an urban classification: a case study in Denver, Colorado. Int. J. Remote Sens. 35 (6), 2067–2081.
Johnston, C.A., 2013. Wetland losses due to row crop expansion in the Dakota Prairie Pothole Region. Wetlands 33 (1), 175–182.
Johnston, C.A., 2014. Agricultural expansion: land use shell game in the US Northern Plains. Landsc. Ecol. 29 (1), 81–95.
Krehbiel, C., Henebry, G.M., 2016. A comparison of multiple datasets for monitoring thermal time in urban areas over the US Upper Midwest. Remote Sens. 8 (4), 297.
Krehbiel, C.P., Jackson, T., Henebry, G.M., 2016. Web-enabled Landsat data time series for monitoring urban heat island impacts on land surface phenology. IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 9 (5), 2043–2050.
Krehbiel, C., Zhang, X., Henebry, G.M., 2017. Impacts of thermal time on land surface phenology in urban areas. Remote Sens. 9 (5), 499.
Lark, T.J., Salmon, J.M., Gibbs, H.K., 2015. Cropland expansion outpaces agricultural and biofuel policies in the United States. Environ. Res. Lett. 10 (4), 044003.
Lark, T.J., Mueller, R.M., Johnson, D.M., Gibbs, H.K., 2017. Measuring land-use and land-cover change using the US Department of Agriculture's cropland data layer: cautions and recommendations. Int. J. Appl. Earth Obs. Geoinf. 62, 224–235.
Liaw, A., Wiener, M., 2002. Classification and regression by randomForest. R News 2 (3), 18–22.
Lloyd, D., 1990. A phenological classification of terrestrial vegetation cover using shortwave vegetation index imagery. Int. J. Remote Sens. 11 (12), 2269–2279.
Lyons, M.B., Keith, D.A., Phinn, S.R., Mason, T.J., Elith, J., 2018. A comparison of resampling methods for remote sensing classification and accuracy assessment. Remote Sens. Environ. 208, 145–153.
Meehan, T.D., Hurlbert, A.H., Gratton, C., 2010. Bird communities in future bioenergy landscapes of the Upper Midwest. Proc. Natl. Acad. Sci. 107 (43), 18533–18538.
Millard, K., Richardson, M., 2013. Wetland mapping with LiDAR derivatives, SAR polarimetric decompositions, and LiDAR–SAR fusion using a random forest classifier. Can. J. Remote. Sens. 39 (4), 290–307.
Millard, K., Richardson, M., 2015. On the importance of training data sample selection in random forest image classification: a case study in peatland ecosystem mapping. Remote Sens. 7 (7), 8489–8515.
Montgomery, D.R., 2007. Soil erosion and agricultural sustainability. Proc. Natl. Acad. Sci. 104 (33), 13268–13272.
Mueller, R., Seffrin, R., 2006. New methods and satellites: a program update on the NASS cropland data layer acreage program. Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. 36, 97–102.
Mutter, M., Pavlacky, D.C., Lanen, N.J., Grenyer, R., 2015. Evaluating the impact of gas extraction infrastructure on the occupancy of sagebrush-obligate songbirds. Ecol. Appl. 25 (5), 1175–1186.
NASA LP DAAC, 2013. MODIS terra/aqua land surface temperature/emissivity 8-day L3 global 1 km SIN grid. Version 5. NASA EOSDIS Land Processes DAAC, USGS Earth Resources Observation and Science (EROS) Center, Sioux Falls, South Dakota. https://lpdaac.usgs.gov, Accessed date: 15 January 2017.
Otto, C.R., Roth, C.L., Carlson, B.L., Smart, M.D., 2016. Land-use change reduces habitat suitability for supporting managed honey bee colonies in the Northern Great Plains. Proc. Natl. Acad. Sci. 113 (37), 10430–10435.
Preston, T.M., Kim, K., 2016. Land cover changes associated with recent energy development in the Williston Basin; Northern Great Plains, USA. Sci. Total Environ. 566, 1511–1518.
Qader, S.H., Dash, J., Atkinson, P.M., Rodriguez-Galiano, V., 2016. Classification of vegetation type in Iraq using satellite-based phenological parameters. IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 9 (1), 414–424.
Rashford, B.S., Walker, J.A., Bastian, C.T., 2011. Economics of grassland conversion to cropland in the Prairie Pothole Region. Conserv. Biol. 25 (2), 276–284.
Reed, B.C., Brown, J.F., Vanderzee, D., Loveland, T.R., Merchant, J.W., Ohlen, D.O., 1994. Measuring phenological variability from satellite imagery. J. Veg. Sci. 5 (5), 703–714.
Reitsma, K.D., Dunn, B.H., Mishra, U., Clay, S.A., Desutter, T., Clay, D.E., 2015. Land-use change impact on soil sustainability in a climate and vegetation transition zone. Agron. J. 107 (6), 2363–2372.
Reitsma, K.D., Clay, D.E., Clay, S.A., Dunn, B.H., Reese, C., 2016. Does the US cropland data layer provide an accurate benchmark for land-use change estimates? Agron. J. 108 (1), 266–272.
Schwartz, M.D., Reed, B.C., 1999. Surface phenology and satellite sensor-derived onset of greenness: an initial comparison. Int. J. Remote Sens. 20 (17), 3451–3457.
Searchinger, T., Heimlich, R., Houghton, R.A., Dong, F., Elobeid, A., Fabiosa, J., ... Yu, T.H., 2008. Use of US croplands for biofuels increases greenhouse gases through emissions from land-use change. Science 319 (5867), 1238–1240.
Singh, N., Kline, K.L., Efroymson, R.A., Bhaduri, B., O'Banion, B., 2017. Uncertainty in estimates of bioenergy-induced land use change: the impact of inconsistent land cover data sets and land class definitions. In: Bioenergey and Land Use Change. 231. pp. 143.
Stephens, S.E., Walker, J.A., Blunck, D.R., Jayaraman, A., Naugle, D.E., Ringelman, J.K., Smith, A.J., 2008. Predicting risk of habitat conversion in native temperate grasslands. Conserv. Biol. 22 (5), 1320–1330.
USDA-FSA (United States Department of Agriculture, Farm Service Agency), 2013. National Agriculture Imagery Program (NAIP). Retrieved from: https://www.fsa.usda.gov/programs-and-services/aerial-photography/index.
USDA-NASS (U.S. Department of Agriculture, National Agriculture Statistics Service), 2018. Cropland data layer metadata. Retrieved from: https://www.nass.usda.gov/Research_and_Science/Cropland/metadata/meta.php.
USGS (U.S. Geological Survey), 2016. Landsat update special issue 4: Landsat Collection 1 level-1 data processing starting soon. Retrieved from: https://landsat.usgs.gov/sites/default/files/documents/LU_Special_Iss4_2016.pdf.
USGS (U.S. Geological Survey), 2017. Product guide: Landsat surface reflectance-derived spectral indices. Retrieved from: https://landsat.usgs.gov/sites/default/files/documents/si_product_guide.pdf.
Vaché, K.B., Eilers, J.M., Santelmann, M.V., 2002. Water quality modeling of alternative agricultural scenarios in the US Corn Belt. J. Am. Water Resour. Assoc. 38 (3), 773–787.
White, M.A., Thornton, P.E., Running, S.W., 1997. A continental phenology model for monitoring vegetation responses to interannual climatic variability. Glob. Biogeochem. Cycles 11 (2), 217–234.
Wimberly, M.C., Narem, D.M., Bauman, P.J., Carlson, B.T., Ahlering, M.A., 2018. Grassland connectivity in fragmented agricultural landscapes of the north-central United States. Biol. Conserv. 217, 121–130.
World Wildlife Fund (WWF), 2016. Plow print report. Retrieved from: http://www.worldwildlife.org/projects/plowprint-report.
Wright, C.K., Wimberly, M.C., 2013a. Recent land use change in the Western Corn Belt threatens grasslands and wetlands. Proc. Natl. Acad. Sci. 110 (10), 4134–4139.
Wright, C.K., Wimberly, M.C., 2013b. Reply to Kline et al.: Cropland data layer provides a valid assessment of recent grassland conversion in the Western Corn Belt. Proc. Natl. Acad. Sci. 110 (31), E2864.
Zhang, X., Friedl, M.A., Schaaf, C.B., Strahler, A.H., Hodges, J.C., Gao, F., ... Huete, A., 2003. Monitoring vegetation phenology using MODIS. Remote Sens. Environ. 84 (3), 471–475.
Zhong, L., Hawkins, T., Biging, G., Gong, P., 2011. A phenology-based approach to map crop types in the San Joaquin Valley, California. Int. J. Remote Sens. 32 (22), 7777–7804.
Zhu, Z., Woodcock, C.E., 2014. Continuous change detection and classification of land cover using all available Landsat data. Remote Sens. Environ. 144, 152–171.